

PE-EC703B — Wireless Sensor Networks

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Module 2 Exam Notes: MANETs, Enabling Technologies & Single-Node Architecture

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1. Mobile Ad-hoc Networks (MANETs)

MANET — Definition

A **Mobile Ad-hoc Network (MANET)** is a self-configuring, infrastructure-free wireless network of mobile devices (laptops, PDAs, smartphones) that communicate directly with each other through multi-hop wireless links. There is no fixed base station or access point; every node acts as both a **host** and a **router**.

1.1 Key Characteristics of MANETs

- **Dynamic topology:** Nodes move freely; links appear and disappear.
- **Self-forming:** No pre-deployed infrastructure needed.
- **Multi-hop routing:** Nodes relay packets for each other to extend range.
- **Bandwidth-constrained:** Shared wireless medium with interference.
- **Energy-constrained:** Relies on batteries of mobile devices.

1.2 MANET Applications

Military field operations, emergency/disaster recovery, vehicle-to-vehicle (V2V) communication, conference/meeting room networking.

2. MANETs vs. Wireless Sensor Networks

Although both MANETs and WSNs use multi-hop wireless communication, they differ fundamentally in design goals, node capabilities, and communication patterns.

Parameter	MANET	WSN
Primary purpose	General-purpose communication between mobile users	Sensing and reporting physical phenomena
Node type	High-capability devices (laptops, smartphones)	Resource-constrained sensor nodes
Node count	Tens to hundreds	Hundreds to thousands
Node mobility	High (users move actively)	Usually static (random deployment)
Energy concern	Moderate (battery replaceable)	Critical (battery often not replaceable)
Communication	Peer-to-peer, bidirectional	Many-to-one (sensors → sink)
Addressing	IP-address based	Data-centric / attribute-based
Data rate	High (Mbps)	Very low (kbps)
Deployment	Planned (users carry nodes)	Unplanned (random scatter)
Failure handling	Rare; disruptive if it occurs	Expected; protocols must tolerate it

3. Enabling Technologies for WSN

The viability of WSN as a technology depends on simultaneous advances in several enabling domains.

Enabling Technologies for WSN

- MEMS (Micro-Electro-Mechanical Systems):** Miniaturizes sensors and actuators to millimeter or sub-millimeter scale. Enables fabrication of temperature, pressure, acceleration, and chemical sensors on a single chip. Low cost, low power, high integration.
- Low-power wireless communication:** Standards like IEEE 802.15.4 (250 kbps, ~1 mW TX power) make radio communication feasible within the energy budget of a small battery. Duty-cycling the radio (sleep/wake) further reduces average power to μW range.
- Low-power embedded processors:** Microcontrollers like TI MSP430, Atmel ATmega, ARM Cortex-M0 operate at μA in sleep mode. Active mode at 1–8 MHz consumes 1–5 mA, enabling months of battery life.
- Energy harvesting:** Supplements or replaces batteries by harvesting ambient energy:
 - Solar:* 10–100 mW/cm² outdoors; ~0.1 mW/cm² indoors.
 - Vibration (piezoelectric):* 0.01–1 mW from mechanical vibrations.
 - Thermal gradient:* Thermoelectric generators (TEGs) using Seebeck effect.
 - RF energy harvesting:* Capture ambient RF signals (very low, μW).

5. **Embedded operating systems:** Lightweight OSes (TinyOS, Contiki, RIOT) designed for <10 KB RAM environments. Event-driven models and component-based architectures reduce code size and power.
6. **Low-power networking protocols:** Cross-layer protocol design (MAC + routing + application co-optimized for energy). IEEE 802.15.4, ZigBee, 6LoWPAN enable mesh networking with minimal overhead.

4. Issues and Challenges in Wireless Sensor Networks

4.1 Hardware-Level Issues

- **Energy scarcity:** Transmission costs $10\times$ – $100\times$ more energy than processing. The radio must be duty-cycled aggressively.
- **Unreliable hardware:** Sensors fail; batteries drain unevenly; node failures are the norm.
- **Limited memory:** Routing tables, buffered data, and OS must fit in <10 KB RAM.

4.2 Network-Level Issues

- **Scalability:** Protocols must work from 10 to 10,000 nodes without reengineering.
- **Dynamic topology:** Node failures and movement change the network graph; routing must adapt.
- **Broadcast nature of wireless:** Collisions waste energy; MAC layer must handle medium access carefully.
- **Hidden terminal problem:** Two nodes out of range of each other both transmit to a common receiver, causing collision.
- **Localization:** GPS is too costly; algorithms like DV-hop, ToA/TDoA, RSSI-based must estimate position.
- **Time synchronization:** Nodes' clocks drift; synchronized sleep schedules and time-stamped data require periodic resynchronization (RBS, FTSP, TPSN protocols).

4.3 Application-Level Issues

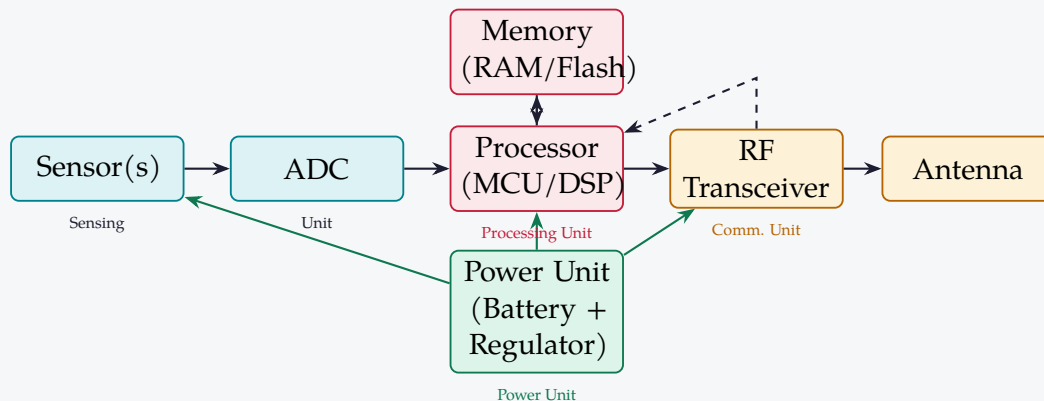
- **QoS requirements:** Different applications need different latency/reliability trade-offs.
- **Data quality:** Sensor readings have noise, drift, and calibration errors.
- **Security and privacy:** Nodes in the field can be captured; data can be eavesdropped or tampered.

5. Single-Node Architecture

Sensor Node Architecture — Definition

A **sensor node** (also called a **mote**) is the fundamental building block of a WSN. It integrates four subsystems on a small PCB: a **sensing unit**, a **processing unit**, a **communication unit**, and a **power unit**. All units are co-designed to minimize energy consumption while meeting the sensing and communication requirements of the application.

Single Sensor Node — Block Diagram



5.1 Subsystems of a Sensor Node

1. Sensing Unit

Consists of one or more physical **sensors** and an **Analog-to-Digital Converter (ADC)**.

- Sensor converts a physical parameter (temperature, light, pressure) into an analog electrical signal.
- ADC digitizes the analog signal for processing. Resolution: typically 8–12 bits. Sampling rate: application-dependent.
- Types of sensors used: resistive (NTC thermistor), capacitive (pressure), piezoelectric (vibration), MEMS accelerometers, optical (photodiode).

2. Processing Unit

A **microcontroller (MCU)** that executes the embedded OS and application firmware.

- Runs sensing schedule, data aggregation, routing protocol, and radio control.
- Includes on-chip **RAM** (2–10 KB) and **Flash** (32–256 KB).
- Examples: TI MSP430 (16-bit, 16 MHz, 165 μ A active); ATmega128L (8-bit); ARM Cortex-M0.

3. Communication Unit

A **wireless radio transceiver** operating in ISM band (typically 2.4 GHz or 868/915 MHz).

- Implements IEEE 802.15.4 physical layer (O-QPSK modulation, 250 kbps).
- The radio is the most energy-hungry component: TX $\approx$ 17 mA, RX $\approx$ 20 mA; idle $\approx$ 1 mA; sleep $< 1 \mu\text{A}$.
- **Key design rule:** Duty-cycle the radio — keep it in sleep mode as long as possible.

4. Power Unit

Provides regulated voltage to all subsystems.

- Primary source: AA or AAA alkaline batteries (1000–3000 mAh), lithium coin cells, or rechargeable LiPo.
- Energy harvesting can supplement: solar panels (thin-film), piezoelectric harvesters, thermoelectric generators.
- A low-dropout (LDO) regulator provides stable 3.3 V (or 1.8 V) rail.

6. Hardware Components and Design Constraints

6.1 Key Hardware Components

Component	Function	Example / Spec
Sensor	Convert physical quantity to electrical signal	NTC thermistor, MEMS accelerometer, CMOS image sensor
ADC	Digitize analog sensor output	12-bit, 100 kSps, built into MCU
Microcontroller	Sensing, processing, routing decisions	TI MSP430, ATmega128L, ARM Cortex-M0
Memory (Flash)	Store firmware, data logs	64–256 KB on-chip; external SPI Flash
Memory (RAM)	Runtime data, routing tables, buffers	2–10 KB on-chip
RF Transceiver	Wireless TX/RX	CC2420, CC2530 (IEEE 802.15.4); nRF24L01
Antenna	Radiate/receive RF signal	Integrated PCB trace, chip antenna, wire monopole
Battery	Primary energy source	AA alkaline 2400 mAh; CR2032 coin 225 mAh
LDO Regulator	Stable voltage rail to all ICs	3.3 V, 300 mA output
Oscillator / RTC	CPU clock and time-keeping	32 kHz crystal for sleep; 8 MHz for active

6.2 Design Constraints

Design of a sensor node is governed by conflicting constraints that require careful trade-offs.

WSN Node Design Constraints

1. **Energy constraint:** Total energy $E = C \times V$. For a 2 AA cell pack (3 V, 2400 mAh) = 7.2 Wh. Every design decision must minimize current draw. **Radio sleep** is non-negotiable. Never keep the radio in RX idle — this wastes as much power as transmitting.
2. **Cost constraint:** Large-scale deployment requires node BOM cost < Rs. 2000. This rules out GPS, high-quality ADCs, or large batteries. COTS low-cost transceivers (CC2520 $\approx$ Rs. 80) are used.
3. **Size constraint:** Nodes must be small enough to embed in targets (soil probes, clothing, walls). PCB area typically < $6 \times 6 \text{ cm}^2$. This limits battery size and antenna length.
4. **Memory constraint:** All code (OS + app + routing) must fit in 64–256 KB Flash. All runtime state in 4–10 KB RAM. Protocols must be algorithmically simple.
5. **Communication range constraint:** IEEE 802.15.4 at 0 dBm TX gives $\sim 30 \text{ m}$ indoor / $\sim 100 \text{ m}$ outdoor range. Multi-hop routing extends coverage.
6. **Reliability/robustness constraint:** Nodes must operate in extreme temperatures (-20°C to $+85^\circ\text{C}$), humidity, and vibration. Industrial-grade MCUs and sealed enclosures are used.

6.3 Energy Consumption Model

The energy consumed by a node has four components:

Node Energy Model

$$E_{\text{node}} = E_{\text{sense}} + E_{\text{process}} + E_{\text{tx}} + E_{\text{rx}}$$

- $E_{\text{sense}} = I_{\text{sensor}} \times V \times t_{\text{sense}}$ (sensor active time)
- $E_{\text{process}} = I_{\text{MCU_active}} \times V \times t_{\text{process}}$
- $E_{\text{tx}} = (E_{\text{elec}} + \varepsilon_{\text{amp}} \cdot d^n) \times l$ (l = bits, d = distance, $n = 2$ or 4)
- $E_{\text{rx}} = E_{\text{elec}} \times l$

Key insight: Communication energy $\gg$ computation energy. Reducing data volume (aggregation) and distance (clustering) drastically extends node lifetime.

7. Quick Revision — Module 2

Key Formulas at a Glance

- Node energy budget: $E_{\text{node}} = E_{\text{sense}} + E_{\text{process}} + E_{\text{tx}} + E_{\text{rx}}$
- TX energy: $E_{tx}(l, d) = l \cdot E_{\text{elec}} + l \cdot \varepsilon_{\text{amp}} \cdot d^n$
- RX energy: $E_{rx}(l) = l \cdot E_{\text{elec}}$
- Node lifetime: $T = \frac{E_{\text{initial}}}{P_{\text{avg}}}$ where P_{avg} = average power draw
- Energy from battery: $E = Q \times V$ (Q = capacity in Ah, V = voltage)
- Solar harvesting power: $P_{\text{solar}} = A \times \eta \times G$ (A = panel area, η = efficiency, G = irradiance)

One-Line Definitions

- **MANET:** Self-organizing mobile wireless network without fixed infrastructure; every node is both host and router.
- **Sensing unit:** Sensor + ADC subsystem that converts a physical quantity into a digital value.
- **Processing unit:** MCU + memory subsystem running the OS, application, and routing firmware.
- **Communication unit:** RF transceiver + antenna enabling wireless data exchange.
- **Power unit:** Battery + regulator providing stable power to all node subsystems.
- **MEMS:** Micro-Electro-Mechanical Systems — miniaturized sensor fabrication technology enabling cheap, tiny transducers.
- **Energy harvesting:** Generating electricity from ambient sources (solar, vibration, thermal) to supplement or replace batteries.
- **Duty cycling:** Periodically turning off the radio (and MCU) to reduce average power; fundamental to WSN energy management.

Frequently Asked / Viva Questions — Module 2

1. Compare MANET and WSN on at least five parameters.

MANET vs WSN: (1) Node capability — MANET uses smartphones/laptops vs WSN uses resource-constrained nodes; (2) Node count — MANET few hundred vs WSN thousands; (3) Energy — MANET moderate vs WSN critical; (4) Communication pattern — MANET peer-to-peer vs WSN many-to-one to sink; (5) Addressing — MANET IP-based vs WSN data-centric; (6) Data rate — MANET Mbps vs WSN kbps.

2. What are the enabling technologies that make WSNs feasible?

Frequently Asked / Viva Questions (contd.)

Six enabling technologies: (1) MEMS for miniaturized cheap sensors; (2) Low-power RF (IEEE 802.15.4) for energy-efficient wireless; (3) Low-power MCUs (MSP430) for μA -level sleep; (4) Energy harvesting (solar, piezo, thermal) for battery-less operation; (5) Lightweight embedded OSes (TinyOS, Contiki); (6) Cross-layer networking protocols (ZigBee, 6LoWPAN).

3. Draw and explain the block diagram of a sensor node.

A sensor node has four subsystems: **Sensing unit** (sensors + ADC converts physical parameters to digital), **Processing unit** (MCU + RAM/Flash runs OS and algorithms), **Communication unit** (RF transceiver + antenna for wireless data), **Power unit** (battery + LDO regulator). All subsystems are co-designed for minimum energy.

4. Why is the communication unit the most energy-hungry component of a sensor node?

The RF radio draws 17–20 mA in TX/RX vs 1–5 mA for the MCU and < 1 mA for sensors. Even in idle listen mode the radio draws ~ 1 mA — equivalent to thousands of MCU sleep cycles. This makes radio duty-cycling (aggressive sleep scheduling) the primary energy optimization technique.

5. What is MEMS and how does it enable WSN?

MEMS (Micro-Electro-Mechanical Systems) uses semiconductor fabrication techniques to build mechanical structures (membranes, cantilevers) alongside electronics on a single chip. This creates cheap ($< \text{Rs. } 50$), tiny, low-power sensors for temperature, pressure, acceleration, and gas detection — prerequisites for mass-deployable WSN nodes.

6. List and briefly explain the six design constraints of a sensor node.

(1) **Energy** — limited non-rechargeable battery; (2) **Cost** — mass deployment, Rs. 100–2000 per node; (3) **Size** — PCB must be compact ($< 6 \times 6$ cm); (4) **Memory** — only KB-level RAM/Flash; (5) **Communication range** — 10–100 m; (6) **Reliability** — must work in harsh environments.

7. What is energy harvesting? Give three examples used in WSN.

Energy harvesting captures ambient energy and converts it into electrical energy to supplement or replace batteries. Examples: (1) **Solar**: Photovoltaic cells convert light to electricity ($10\text{--}100 \text{ mW/cm}^2$ outdoor); (2) **Piezoelectric**: Mechanical vibrations deform piezo material generating voltage; (3) **Thermoelectric**: Seebeck effect converts temperature gradient across a thermoelectric module into voltage.

8. What is the hidden terminal problem in WSNs?

Nodes A and C are both in range of B but not of each other. When A and C simultaneously transmit to B, their signals collide at B. Neither A nor C detects the collision (they cannot hear each other). This is the hidden terminal problem, solved using RTS/CTS handshaking or CSMA with virtual carrier sensing.

9. What is duty cycling and why is it critical in WSN?

Frequently Asked / Viva Questions (contd.)

Duty cycling = fraction of time the radio (or MCU) is ON. If a node transmits 1 ms every 1 s, duty cycle = 0.1%. If idle-listen current = 20 mA and duty cycle = 1%, average current ≈ 0.2 mA — extending a 2000 mAh battery from ~ 4 days (always-on) to ~ 400 days.

10. How does a sensor convert a physical quantity to digital data?

The physical quantity (e.g., temperature) changes the sensor's electrical property (e.g., NTC thermistor resistance). This is connected in a voltage divider; the analog voltage is fed to an ADC (Analog-to-Digital Converter). The ADC samples the voltage and quantizes it to n -bit digital code (e.g., 12-bit = 4096 levels) for the MCU.

11. Name two lightweight embedded OSES used in WSN and one property each.

(1) **TinyOS**: Event-driven, component-based OS developed at UC Berkeley for motes; fits in < 4 KB RAM. (2) **Contiki**: Protothreads-based OS supporting IPv6/6LoWPAN; enables internet connectivity for sensor nodes.

12. What is the significance of the radio transceiver's idle-listen state?

In idle-listen, the radio is powered ON and listening but no data is being received. Current draw is similar to receive mode (~ 20 mA). A protocol that leaves the radio always-on in idle-listen wastes enormous energy even with no traffic. This motivates sleep-wake scheduling MAC protocols (S-MAC, TDMA) that turn off the radio between duty cycles.

Summary: MANET vs WSN at a Glance

Attribute	MANET	WSN
Node capability	High	Very low
Energy priority	Moderate	Critical
Traffic pattern	Peer-to-peer	Many-to-one
Addressing	IP-based	Data-centric
Deployment	Planned	Random
Node failures	Exceptional	Expected